Chapter 6 Notes

Section 1

Notes

Note that $F \upharpoonright A$ does not necessarily equal $A \times F''(A)$. In fact, if $F''(A)$ contains more than one element (say it contains at least the two elements b, c), then obviously $F \upharpoonright A \neq A \times F''(A)$, as, for any element $x \in A$, we would have $\langle x, b\rangle$ and $\langle x, c\rangle$ both in $A \times F''(A)$, but there can only be one y for which $\langle x, y\rangle \in F \upharpoonright A$.

It is remarked that:"If ϕ_1 and ϕ_2 are functions from a class A onto a class B and if $\phi_1\phi_2^{-1}$ is an identity function, then obviously $\phi_1 = \phi_2$." Here is why this is. Now, note that this assumes that ϕ_2^{-1}, is, in fact, a function. Then, for every element y in the range of ϕ_2, we have $\phi_2^{-1}(y) = x$ for some x in the domain of ϕ_2, and $\phi_1\phi_2^{-1}(y) = \phi_1(x) = y$. But also, $\phi_2(x) = y$. Thus $\phi_1(x) = \phi_2(x)$ for every x in the domain of ϕ_2, and because they have the same domains, the functions are the same.

Note that if a class A is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. The proof of this is the same as the proof of Proposition 1.3.

Exercise 1.1

Let $\mathcal{N}$ be a nest of functions.

(1): We first show that $\cup\mathcal{N}$ is function. Obviously $\cup\mathcal{N}$ is a relation. Now, suppose we had $\langle x, y\rangle$ and $\langle x, z\rangle$ both in $\cup\mathcal{N}$. Thus, for some functions f and g, $\langle x, y\rangle \in f$ and $\langle x, z\rangle \in g$. But, because $\mathcal{N}$ is a nest, either $f \subseteq g$ or $g \subseteq f$. Without loss of generality, suppose $f \subseteq g$. Then we would have $g(x) = y$ and $g(x) = z$, and because g is a function, $y = z$. Hence $\cup\mathcal{N}$ must be a function.

(2): We now show that the domain of $\cup\mathcal{N}$ is the union of the class of domains of the elements of $\mathcal{N}$ (call this D). Suppose $x \in \mathrm{Dom}(\cup\mathcal{N})$. Then, for some y and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in D$. On the other hand, suppose $x \in D$. Then, for some y and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in \mathrm{Dom}(\cup\mathcal{N})$. Thus $\mathrm{Dom}(\cup\mathcal{N}) = D$. Similar reasoning shows $\mathrm{Ran}(\cup\mathcal{N})$ is the union of the class of ranges of the elements of $\mathcal{N}$.

(3): Lastly, suppose that every element of $\mathcal{N}$ was 1-1. Now, suppose x and y are in the domain of $\cup\mathcal{N}$, and that $x \neq y$, and for some v and w, $\langle x, v\rangle \in \cup\mathcal{N}$ and $\langle y, w\rangle \in \cup\mathcal{N}$. Then, for some functions f and g in $\mathcal{N}$, $\langle x, v\rangle \in f$ and $\langle y, w\rangle \in g$. Without loss of generality, suppose that $f \subseteq g$. Then g is 1-1, meaning $v \neq w$. Hence $\cup\mathcal{N}$ is 1-1.

Exercise 1.2

First, let us note that On is not a set. For if it were, it would then be a transitive set of ordinals, and thus would be an ordinal. Then that would mean $On \in On$, which never holds for ordinals. Thus On is not a set.

Now, let A be the class $On \cup \{\{\{\emptyset\}\}\}$. Thus $x \in A$ iff either x is an ordinal, or $x = \{\{\emptyset\}\}$. Note that $\{\{\emptyset\}\}$ is not an ordinal, as if it were, then it would be transitive, which would imply that $\emptyset \in \{\{\emptyset\}\}$, which is not the case. Hence On is a proper subclass of A.

Then define the relation $x \leq y$ (on A) as follows: $x \leq y$ iff x and y are both ordinals and $x \subseteq y$, or $y = \{\{\emptyset\}\}$. Note that this defintion implies that $\{\{\emptyset\}\} \leq y$ only holds if $y = \{\{\emptyset\}\}$. We first show that $\leq$ is a well ordering of A.

O_1: If x is an ordinal, then $x \subseteq x$, so that $x \leq x$ holds. If $x = \{\{\emptyset\}\}$, then $x \leq x$ by definition.

O_2: Suppose $(x \leq y) \wedge (y \leq z)$. If all of x, y and z are ordinals, then obviously $x \leq z$. If any of x, y and z were equal to $\{\{\emptyset\}\}$, then $z = \{\{\emptyset\}\}$ must hold, which of course implies that $x \leq z$.

O_3: Suppose $x \leq y$ and $y \leq x$. Then either $x = y = \{\{\emptyset\}\}$, or both x and y are ordinals, in which case $x \subseteq y$ and $y \subseteq x$, so that $x = y$. Thus $x = y$.

O_4: If x and y are both ordinals, then of course either $x \leq y$ or $y \leq x$. If $y = \{\{\emptyset\}\}$, then $x \leq y$ (similarly with x).

O_5: For any subclass B of A that only contains ordinals, then, because On is well ordered under $\subseteq$, it contains a least element. If $\{\{\emptyset\}\} \in B$, then $B - \{\{\{\emptyset\}\}\}$ contains a least element e, and e is also the least element of B itself, as $e < \{\{\emptyset\}\}$. Thus every subclass B of A contains a least element.

Thus $\leq$ is a well ordering of A. Now, On is a proper lower section of A: It is a proper subclass of A, and $x < \{\{\emptyset\}\}$ for all elements in On. But On is not a set. Thus we have an example of a well ordering which is not a proper well ordering.

Section 2

Notes

Note that any subclass of an isomorphism from A onto B is an isomorphism from a subclass of A onto a subclass of B.

Obviously, if A is isomorphic to B, then B is isomorphic to A (as seen in the next paragraph).

Note that, if ϕ is an isomorphism from A onto B, then ϕ^{-1} is an isomorphism from B onto A. For, clearly ϕ^{-1} is a 1-1 function from B onto A. And suppose ϕ is an isomorphism from the system $\langle A, \leq_1 \rangle$ to the system $\langle B, \leq_2 \rangle$. Then, for any a_1, a_2 in B, $a_1 <_2 a_2$ iff $\phi(a_1) <_1 \phi(a_2)$.

The above fact is used, for instance, in Corollary 2.2. Also, in Corollary 2.2, note that $(2')$ follows because $\phi(\phi^{-1}(a)) = a$.

I think there is an error in the proof of Corollary 2.3. It claims that $\phi_1\phi_2^{-1}$ is an isomorphism from A onto A, but this is clearly not the case. For ϕ_2^{-1} is from B onto A, and ϕ_1 is from A onto B, meaning $\phi_1\phi_2^{-1}$ is actually from B onto B (now, this would also work, as then, by Corollary 2.2, $\phi_1 = \phi_2$, but it is explicitly said in the proof that $\phi_1\phi_2^{-1}$ is from A onto A). If we wanted to get a function from A onto A, we would need to pick $\phi_2^{-1}\phi_1$ (for example; there are others).

Here is why $\phi_1\phi_2^{-1}$ is in fact an isomorphism. For, if a_1 precedes a_2 in B, then $\phi_2^{-1}(a_1)$ precedes $\phi_2^{-1}(a_2)$ in A, and $\phi_1(\phi_2^{-1}(a_1))$ precedes $\phi_1(\phi_2^{-1}(a_2))$ in B (more generally, if f is an isomorphism from A onto B, and g is an isomorphism from B onto A, then fg is an isomorphism from B onto B).

Also, here is why (as said in Corollary 2.3), if $\phi_1\phi_2^{-1}$ is the identity map (for ϕ_1 and ϕ_2 both from A onto B), then $\phi_1 = \phi_2$. For suppose that $\phi_1\phi_2^{-1}$ is the identity map. Then, for any $x \in B$, $\phi_2^{-1}(x) = y$ for some $y \in A$, and $\phi_1(y) = x$, but also $\phi_2(y) = x$, meaning $\phi_1 = \phi_2$.

Corollary 2.5 holds, as, for any two distinct ordinals α, β, they are comparable. Without loss of generality, suppose $\alpha < \beta$. Now, α must be an initial segment of β (clearly, all elements of α are in β, due to the transitivity of ordinals). This is because, for any element $x \in \alpha$, and for any element $y \in \beta$ but not in α, $x < y$ holds because $x < \alpha$ (by assumption), and, if $x \geq y$, then $y < \alpha$. But, this would mean $y \in \alpha$, a contradiction. Hence α is an initial segment of β, and thus there can be no isomorphism from β to α.

Corollary 2.6 holds, as, between two distinct lower sections of the same well ordered system, one is an initial segment of the other. For supppose the two lower sections were $L_<(x)$ and $L_<(y)$. Now, without loss of generality, suppose $x < y$

(and thus $L_<(x)$ is a subset of $L_<(y)$). Then $L_<(x)$ is an initial segment of $L_<(y)$, as, for every $z \in L_<(x)$, obviously $z < w$ for every $w \in L_<(y)$ but not in $L_<(x)$ (for if not, then $z \geq x$), meaning $L_<(x)$ is proper lower section of $L_<(y)$, and hence also an initial segment of $L_<(y)$. Thus, if two distinct lower sections of the same well ordered system are isomorphic to each other, then one of them would be isomorphic to one of its initial segments, contrary to Corollary 2.4.

If A is isomorphic to B, and B isomorphic to C, then A is obviously isomorphic to C. For suppose A is isomorphic to B under ϕ_1, and B is isomorphic to C under ϕ_2. Then let ϕ_3 be the class of all orderd pairs $\langle x, \phi_2(\phi_1(x))\rangle$ where $x \in A$; ϕ_3 is the isomorphism from A to C.

This explains Corollary 2.7: For suppose a lower section of A was isomorphic to two distinct lower sections C and D of B. Then it would follow that A is isomorphic to C, and A is isomorphic to D, meaning C is isomorphic to D. But this would contradict Corollary 2.6.

In Corollary 2.8, here is why $I_2 \upharpoonright D_1$ is an isomorphism from D_1 onto a lower section of B: Clearly $I_2 \upharpoonright D_1$ is an isomorphism from D_1 onto a subclass of the range of I_2. Also, suppose that A was well ordered under $\leq_1$ and B was well ordered under $\leq_2$. Now, suppose C is the range of I_2, and C' is the range of $I_2 \upharpoonright D_1$. Suppose that C' was not a lower section of C. Then, for some $x \in C'$, there would be some $y \in C - C'$ such that $y <_2 x$. But, because I_2 is an isomorphism, $I_2^{-1}(y) <_1 I_2^{-1}(x)$. But $I_2^{-1}(x) \in D_1$, and it was assumed that $D_1 \subseteq D_2$, meaning $I_2^{-1}(x) <_1 I_2^{-1}(y)$ (because D_1 and D_2 are both lower sections of A), meaning it cannot be that $y <_2 x$, and hence $I_2 \upharpoonright D_1$ must be an isomorphism onto C', a lower section of B (that is, a lower section of C, but because C is a lower section of B, then C' is also a lower section of B).

Here is why $\mathcal{I}$ in the proof of Theorem 2.9 must be an isomorphism from a lower section of A onto a lower section of B. Recall that $\mathcal{I} = \cup\mathcal{N}$, and $\mathcal{N}$ is the class of all isomorphisms from lower sections of A which are sets onto lower sections of B which are sets. By Proposition 1.4, $\mathcal{I}$ is a function. The domain of $\mathcal{I}$ must be a lower section of A. For suppose it weren't. Then for some x in the domain of $\mathcal{I}$, there would be some $y \in A$ not in the domain of $\mathcal{I}$ such that $x > y$. But x is also in the domain of some $f \in \mathcal{N}$ which is an isomorphism from a lower section of A, which would mean that y is in the domain of f (and thus also of $\cup\mathcal{N}$), a contradiction. Hence the domain of $\mathcal{I}$ is a lower section of A. Similarly, it can be seen that $\mathcal{I}$ is onto a lower section of B.

Section 3

Exercise 3.1

(a): On the one hand, suppose the axiom of substitution. Now, suppose F is a function and x is a set. Then $F \upharpoonright x$ is a subclass of $x \times F''(x)$. But $F''(x)$ is a set by the axiom of substitution, meaning $x \times F''(x)$ is a set, and hence so is $F \upharpoonright x$. On the other hand, suppose that for any function F and any set x, the function $F \upharpoonright x$ is a set. Then consider any function F and any set x. Note that $F''(x)$ is a subclass of $\cup(F \upharpoonright x)$. Thus, because $F \upharpoonright x$ is a set, so is $\cup(F \upharpoonright x)$, and hence so is $F''(x)$.

(b): On the one hand, suppose the axiom of substitution. Now consider some function F whose domain D is a set. Then $F''(D) = \mathrm{Ran}(F)$. Because $F''(D)$ is a set, then, because $D \times F''(D)$ is a set, and F is a subclass of $D \times F''(D)$, F is a set. On the other hand, suppose that any function whose domain is a set is also a set. Now consider any function F and any set x. Then $F''(x)$ is a function whose domain is x, which is a set, and thus $F''(x)$ is also a set by assumption.

Exercise 3.2

Suppose the axiom of choice, and that any class that can be put into a 1-1 correspondence with a set is a set. Now consider any function F and any set x. Let C be the choice function for x. For each $y \in F''(x)$, let $S(y)$ be the set of all $z \in x$ such that $F(z) = y$ (and $S(y)$ is a set, as it is a subset of x, which is a set by assumption). Then $C(S(y)) \in S(y)$, that is, $C(S(y))$ is some $z \in x$ such that $F(z) = y$. Then let f be the set of all ordered pairs $\langle y, C(S(y))\rangle$ for $y \in F''(x)$. This is a 1-1 correspondence between $F''(x)$ and the set of all elements $z \in x$ such that $z = C(S(y))$ for some $y \in F''(x)$. Hence, because $F''(x)$ can be put into a 1-1 correspondence with a set, it is a set.

Section 4

Notes

Note that every well ordered set is also properly well ordered, because obviously every proper lower section of the well ordered set would be a subset of the set, and hence also a set.

Section 5

Note that, if a class M is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. For consider any proper lower section L of M. Then it is also an initial segment of x for some $x \in M$. Hence, for every $y \in L$, $y \subseteq x$, meaning $L \subseteq \mathcal{P}(x)$, and thus L is a set.

The above explains why M (as defined on page 84) is not only well ordered, but properly well ordered.

Note that, for Theorem 5.1, we of course do not necessarily need that M be minimally superinductive, as long as M is both a set and well orderd under $\subseteq$ such that the three conditions of Theorem 3.3 of Chapter 4 hold.

Here is why, on page 84, if λ is a limit ordinal, then M_λ must be a limit element of M. This follows by the definition of the isomorphism: It cannot be a least element (or else λ would be the least ordinal), and it cannot be a successor element (or else λ would be a successor ordinal). Then, here is why this means that M_λ is the set of all M_α, where $\alpha < \lambda$. For, if $\alpha < \lambda$, then $M_\alpha < M_\lambda$, and hence $M_\alpha \subseteq M_\lambda$. On the other hand, if $x \in M_\lambda$, then $x \in M$ for some $M < M_\lambda$, and it must be that $M = M_\alpha$ for some $\alpha < \lambda$ (or else $M \geq M_\lambda$).

On a side note, it seems to me that, in defining the separation principle, the book really meant to allow $\in$ as the only relation (which would, for instance, be capable of defining $\subseteq$ and $=$ with set quantification). This seems to work, as all sets and classes defined with that principle thus far would only need the relation $\in$ for the definition.

There is a discrepancy in the definitions of an extending operation. The first only says that E must be defined on all ordinal sequences, and that if applied to a sequence of length α, it yields one of length α^+. However, the second definition specifies that E must yield, from a sequence θ_1 of length α, a sequence θ_2 of length α^+ such that $\theta_1 \subseteq \theta_2$.

It should be noted that (with the axiom of substitution), every ordinal sequence is a set. Obviously the domain of every ordinal sequence is a set (because the domain is just an ordinal). Thus the range of every ordinal sequence can be put into a 1-1 correspondence with a set, meaning, by the axiom of substitution, the range is also a set. Then, for every ordinal sequence θ, $\theta \subseteq \mathrm{Dom}(\theta) \times \mathrm{Ran}(\theta)$, which is a set. This explains why we can speak of functions whose domain includes all ordinal sequences, as, because all ordinal sequences are sets, we can quantify over them (in the separation princple).

Uniqueness of Functions

Here is why the function in Theorem 5.3 is unique. For suppose we had two functions F and G satisfying the desired conditions. We show by transfinite induction that $F = G$. The case for 0 follows in this way: $F \restriction 1 = E(F \restriction 0) = E(0)$. Similarly $G \restriction 1 = E(0)$. Thus $F \restriction 1 = G \restriction 1$. But $F \restriction 1 = \langle 0, F(0)\rangle$ and $G \restriction 1 = \langle 0, G(0)\rangle$, and thus $F(0) = G(0)$. Now suppose that, for some α, $F(\beta) = G(\beta)$ for all $\beta < \alpha$. Then $F \restriction \alpha^+ = E(F \restriction \alpha)$ and $G \restriction \alpha^+ = E(G \restriction \alpha)$. But, by assumption, $F \restriction \alpha = G \restriction \alpha$, meaning $F \restriction \alpha^+ = G \restriction \alpha^+$. Hence $\langle \alpha, F(\alpha)\rangle = \langle \alpha, G(\alpha)\rangle$, meaning $F(\alpha) = G(\alpha)$. Thus, by transfinite induction, $F = G$.

Here is why the function in Theorem 5.5 is unqiue. For assume the hypotheses of Theorem 5.5, and that we had two functions F and G such that $F(\alpha) = g(F \restriction \alpha)$ and $G(\alpha) = g(G \restriction \alpha)$ such that the domains of F and G are both On. Then we use transfinite induction to show that $F = G$. The case for the least element holds, as $F(0) = g(F \restriction 0) = g(0) = g(G \restriction 0) = G(0)$. Now suppose that for all ordinals $\beta < \alpha$, $F(\beta) = G(\beta)$. Then $F(\alpha) = g(F \restriction \alpha)$ and $G(\alpha) = g(G \restriction \alpha)$. But, by definition of α, the domain of $F \restriction \alpha$ is every β such that $\beta < \alpha$. Hence, by the assumption that $F(\beta) = G(\beta)$ for every $\beta < \alpha$, $F \restriction \alpha = G \restriction \alpha$, and thus $F(\alpha) = G(\alpha)$. By transfinite induction, it follows that $F(\alpha) = G(\alpha)$ for all α, meaning $F = G$.

Here is the argument showing that the On-sequence in Theorem 5.7 is unique: Suppose we had two On functions F and G satisfying the required conditions. Obviously $F(0) = G(0)$. If $F(\alpha) = G(\alpha)$, then $F(\alpha^+) = g(F(\alpha)) = g(G(\alpha)) = G(\alpha^+)$. If $F(\alpha) = G(\alpha)$ for every $\alpha < \lambda$ for a limit ordinal λ, then $F(\lambda) = \cup_{\alpha<\lambda} F(\alpha) = \cup_{\alpha<\lambda} G(\alpha) = G(\lambda)$. Thus, by transfinite induction, $F = G$.

Lemma 5.2

(1): First, consider some chain C of ordinal sequences. Then $\cup C$ is a function (by Proposition 1.4) and a set. Thus the domain D of $\cup C$ is a set. Now suppose $\alpha \in D$ and $\beta \in \alpha$. Then, for some $f \in C$, $\alpha \in C$. But $\beta < \alpha$, and by definition of f, it follows that $\beta \in f$ (and thus $\beta \in D$). Hence D is transitive, meaning, because it is a transitive set of ordinals, it is also some ordinal α. Then $\cup C$ is an ordinal sequence of length α.

Suppose that λ is a limit ordinal such that all elements of C are of length less than λ, and if for every $\alpha < \lambda$ there is a an element of C of length α. We need to show that $\cup C$ is of length λ, that is, $\alpha \in \mathrm{Dom}(\cup C)$ iff $\alpha < \lambda$. On the one hand, suppose that $\alpha \in \mathrm{Dom}(\cup C)$. Then $\alpha \in f$ for some $f \in C$. By definition of C, f is of a length less than λ, meaning $\alpha < \lambda$. On the other hand, suppose that $\alpha < \lambda$. Then there is an element of C length α. Then there also must be an element f of C of length α^+, as $\alpha^+ < \lambda$ (or else λ would not be a limit element), and $\alpha \in f$, that is, $\alpha \in \mathrm{Dom}(\cup C)$. Thus $\cup C$ is of length λ.

(2): Suppose that $\mathcal{N}$ is a nest of ordinal sequences and that $\mathcal{N}$ is not a set. Again, by Proposition 1.4, $\cup\mathcal{N}$ is a function. First, we show that $\mathcal{N}$ is properly well ordered under $\subseteq$. Because it is a nest, it is linearly ordered under $\subseteq$. Now consider any subclass C of $\mathcal{N}$. Then consider the class F of ordinals α such that a function of length α is in C. Then F has a least element β. Then the function f which is of length β is clearly the least element of C. Hence $\mathcal{N}$ is well ordered under $\subseteq$ (because the well ordering is under $\subseteq$, it is also properly well ordered under $\subseteq$). Then, because $\mathcal{N}$ is not a set, by the counting theorem, there is an isomorphism from $\mathcal{N}$ onto On. Now consider the class A of all ordinals α such that a function f of length α is in $\mathcal{N}$. Because $\mathcal{N}$ is a nest, there can be at most one $f \in \mathcal{N}$ of length α. Hence A can be put into 1-1 correspondence with $\mathcal{N}$, which in turn can be put into a 1-1 correspondence with On. Hence A, a class of ordinals, can be put into a 1-1 corresponence with On. Thus A is a transitive class of ordinals (for, if $\alpha \in \mathrm{Dom}(\cup\mathcal{N})$ and $\beta \in \alpha$, then $\alpha \in f$ for some $f \in \mathcal{N}$ and $\beta < \alpha$, meaning $\beta \in f$) which is not a set (by the axiom of substitution), and is therefore equal to On. Hence, for every α, there is a function of length α in $\mathcal{N}$, which obviously implies that the domain of $\cup\mathcal{N}$ is On.

Theorem 5.3

Here is why we can conclude that, if λ is a limit ordinal and if for each $\alpha < \lambda$, M_α is an α-sequence, then M_λ is a λ sequence. First, note that, by Theorem 5.1, $M_\lambda = \cup_{\alpha<\lambda} M_\alpha$. Thus M_λ is the union of a chain of ordinal sequence (namely, the chain of all ordinal sequence of length $\alpha < \lambda$ where the sequence is in M). Clearly all elements of C are of length less than λ. It remains to show (in order to use part (1) of Lemma 5.2) that for every $\alpha < \lambda$ there is an element of C of length α. This can be proven by transfinite induction on the ordinals: we show that if $\alpha < \lambda$, then there is an ordinal sequence of length α in C. This holds for the 0, as by definition of M, $0 \in M$ (and thus also in C). For any α, if $\alpha < \lambda$ and there is some ordinal sequence of length α in M, then, if $\alpha^+ < \lambda$, by definition of M, there is an ordinal sequence of length α^+ in M. If for all $\alpha < \lambda_1$ for some limit ordinal λ_1, there is some ordinal sequence of length α in M, then, if $\lambda_1 < \lambda$, by definition of M, there is an ordinal sequence of length λ_1 in M. Thus, for every $\alpha < \lambda$, there is an element of C of length α, meaning $\cup C$, that is M_λ, is of length λ.

Theorem 5.8

The proof of this theorem is incorrect (and the theorem itself is incorrect). Now, the proof says:"By Theorem 5.5, taking $g(x) = h(\mathrm{Ran}(x))$, if x is a function (otherwise take $g(x) = x$)." However, this is not correct, as it very well may be that h is not defined for $\mathrm{Ran}(x)$ for some function x. A counter example to this theorem is simply the function $f = \{\langle 1, 1\rangle\}$. For, by the Theorem, we would have a function F such that $F(0) = f(F''(0)) = f(0)$, but 0 is not defined for f.